

Carbon buckyball acts as a wheel, allowing car to roll across a surface of gold.

Nanomachines

Materials such as graphene (see p.237) and buckyballs (see p.222) can be used to build working machines at the nanoscale. In 2005, researchers at Rice University, Texas, made a nanocar out of polymer and carbon molecules. Measuring under 4 nm, the car moved when the surface it rested on was heated above 392°F (200°C).



National Geographic Kids magazine cover is just 11 micrometers (11,000 nm) wide, magnified to about 4,400 times here

Nanopatterning

This process uses a tiny probe around 100,000 times smaller than a pencil point. It is heated to around 1,830°F (1,000°C) to melt a sheet of plastic polymer in order to sculpt a design. In 2014, nanopatterning produced the world's smallest magazine cover (left), small enough for 2,000 copies of the cover to fit on a grain of salt. Future computer parts may be produced using this technique.

Nanobot carries drug in reservoir, which it injects into cell via a retractable needle.

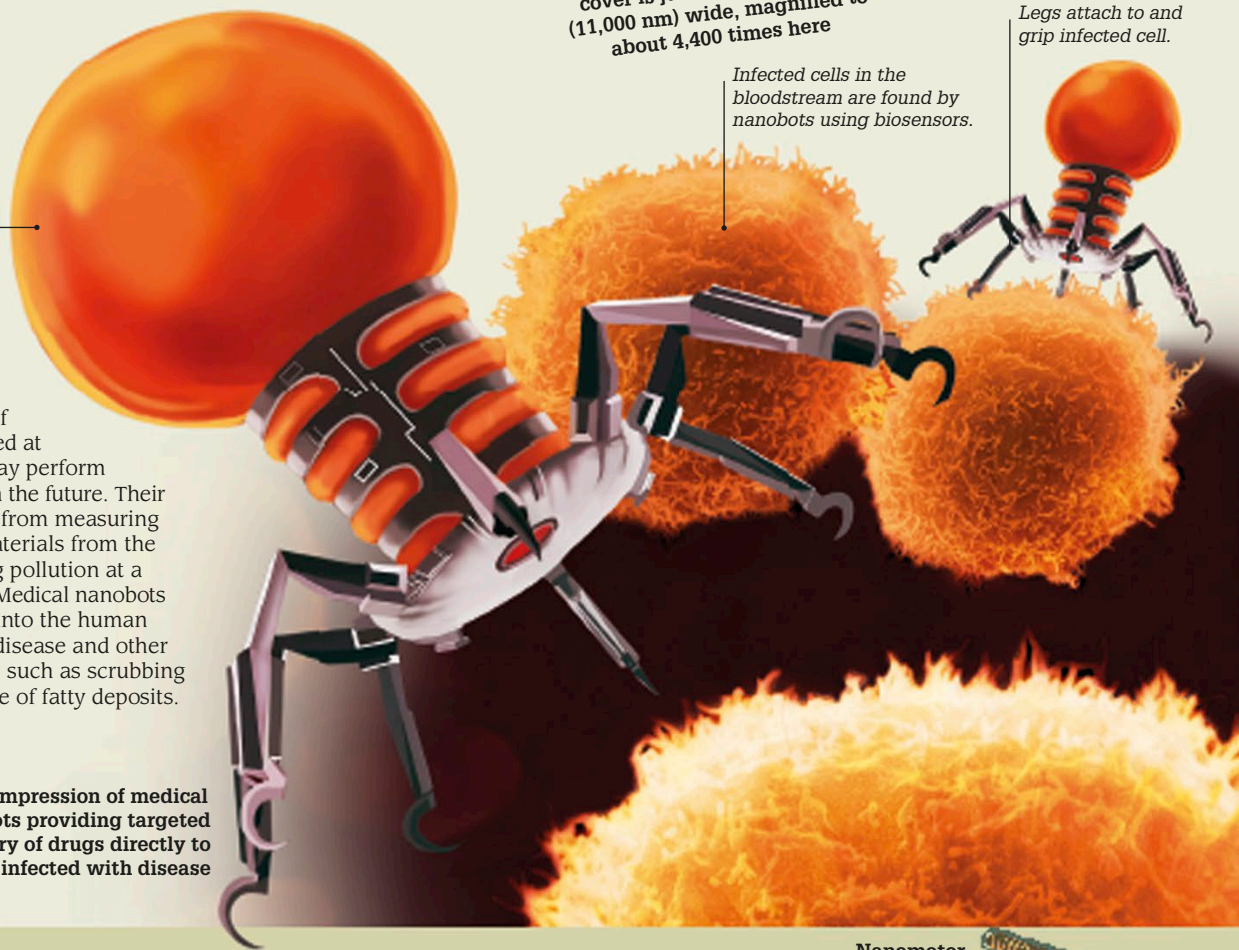
Nanobots

Large numbers of robots constructed at the nanoscale may perform invaluable jobs in the future. Their tasks may range from measuring and repairing materials from the inside or tackling pollution at a molecular level. Medical nanobots may be injected into the human body to combat disease and other health problems, such as scrubbing blood vessels free of fatty deposits.

Artist's impression of medical nanobots providing targeted delivery of drugs directly to cells infected with disease

Infected cells in the bloodstream are found by nanobots using biosensors.

Legs attach to and grip infected cell.



Nanomotor

2003

The world's smallest guitar was made by scientists at Cornell University. Its strings measure 150–200 nm in width and vibrate at frequencies 130,000 times higher than a real guitar.

2008

French physicist Albert Fert and German physicist Peter Grünberg won the Nobel Prize for using metal layers a few atoms thick to discover giant magnetoresistance (GMR). GMR is used to build very high capacity hard disk drives.

2013

Harvard and Illinois University researchers produced batteries smaller than 0.04 in (1 mm) using 3-D printing. Even tinier batteries may be crucial in powering devices such as nanobots.

2014

A nanomotor small enough to fit inside a single human cell was built by a University of Texas team led by American inventor Dr. Donglei Fan (see p.243).



Nanoguitar